

Multi-UAV Trajectory Planning and Task Allocation for Disaster Rescue Operations

Yousra M. Aljifri*, Sarah H. Shaaban*, Farida B. El-Saadawy*, Yousef H. Ali*,
Youssef M. Awad*, Youssef M. El-Khawanky*, Dalia M. Mahfouz[†] and Omar M. Shehata[†]
*Mechatronics Dept., Faculty of Engineering and Materials Science (EMS), German University in Cairo, Egypt
[†]Multi-Robot Systems (MRS) Research Group, Cairo, Egypt
{yousra.aljifri, sarah.shaaban, farida.elsaadawy, yousef.selim, youssef.sayed, youssef.khawanky}@student.guc.edu.eg
{dalia.mahfouz, omarmohamad}@guc.edu.eg

Abstract—This paper presents a constrained multi-UAV routing framework for disaster rescue missions that incorporates load-dependent endurance coupling and time-window constraints within a unified optimization model. Segment cost is dynamically linked to remaining payload, capturing realistic energy–mass interactions during flight. The objective integrates travel cost, UAV activation cost, temporal penalties, and utilization efficiency rewards. Two solution strategies are proposed: a Discrete Teaching–Learning–Based Optimization with Local Search (DTLBO-II-LS), featuring class-based learning and embedded local search operators for robust low-variance combinatorial routing, and a policy-gradient Reinforcement Learning formulation (REINFORCE) that generates adaptive routing policies through episodic sampling without predefined neighborhood operators. Both are benchmarked against three classical metaheuristics: Simulated Annealing (SA), Genetic Algorithm (GA), and Particle Swarm Optimization (PSO) under a unified framework. Across 10- and 20-node scenarios, DTLBO-II-LS attains the lowest variance (4.64 over 20 runs) and outperforms GA by approximately 2% in larger instances, demonstrating superior robustness and convergence stability, while SA achieves the best single-run cost (−950.41). REINFORCE produces competitive feasible solutions and demonstrates adaptive behavior, though variability increases with problem scale. These results demonstrate that structured discrete metaheuristics (SA, GA, DTLBO-II-LS) offer superior solution quality and robustness for time-critical UAV disaster response, while learning-based routing (REINFORCE) provides a flexible, operator-free alternative whose scalability can be improved through enhanced policy parameterization and reward design. The two paradigms thus represent complementary directions: metaheuristics for deployment-ready optimization and RL for adaptive long-term learning.

I. INTRODUCTION

Unmanned aerial vehicles (UAVs) have become an important technology for disaster management and search-and-rescue (SAR) operations due to their rapid deployability, scalability, and ability to operate in hazardous or inaccessible environments [1]. Natural disasters have caused increasing losses worldwide, with material damages rising by approximately 100–150% over the past 30 years [2], and the first 72 hours after a disaster are considered the most critical period for conducting search-and-rescue operations and preserving human lives [2]. In such time-critical situations, UAVs can deliver medical supplies and emergency aid to multiple rescue locations while reducing human exposure and accelerating response time. However, efficient coordination of multiple UAVs remains a challenging problem: a mission planner must determine which UAVs should be deployed, how rescue points should

be assigned, and which routes should be followed, while satisfying operational constraints such as payload capacity, energy limitations, and service time windows.

Recent research has explored various optimization approaches for coordinating UAV fleets in search-and-rescue missions. Han et al. [3] formulated a Rescue Mission UAV Routing Problem with Time Windows (RMURPTW) for post-earthquake SAR operations and proposed a hybrid Particle Swarm Optimization–Grey Wolf Optimizer (PSOGWO) algorithm, demonstrating faster convergence and lower mission cost than the individual algorithms, although the reliance on static parameters limited scalability for larger scenarios. Dong et al. [4] investigated multi-objective optimization for maritime search-and-rescue (MSAR) missions using a Deb-rule enhanced Non-Dominated Sorting Genetic Algorithm II (DNSGA-II), achieving a 6.15% reduction in operational cost and an 11.32% reduction in response time, but with performance degrading under uncertain demand conditions. Cao et al. [5] proposed a robust multi-UAV task allocation model using a hybrid Genetic Algorithm (GA) and Artificial Fish Swarm Algorithm (AFSA), improving convergence and mission cost over conventional GA methods, though the approach required manual parameter tuning and simplified environmental assumptions. Li et al. [6] addressed time-optimal UAV task allocation with battery constraints using a mutation-based Beetle Antennae Search (MBAS) algorithm, reducing mission completion time by up to 20% without considering dynamic environmental conditions. Yang et al. [7] introduced a dual-layer framework combining UAV coverage optimization with human rescue path planning through a clustering-based coverage path optimization algorithm, reducing task completion time by 27.8% under static-map and simplified-communication assumptions. Beyond single-algorithm comparisons, decentralized multi-agent path planning [8] and hybrid decentralized AGV architectures [9] demonstrate scalable routing strategies transferable to aerial fleets, while the broader UAV routing problem landscape surveyed in [10] highlights the need for unified comparative benchmarks.

Overall, existing formulations often assume fixed energy consumption without explicitly coupling UAV payload with endurance, and scalability under statistical robustness analysis remains insufficiently addressed. Learning-based methods are emerging but their applicability to constrained combinatorial routing under hard time windows is still limited. These gaps motivate a unified load-aware framework

with systematic multi-solver comparison.

The contributions of this work are threefold. First, we formulate a multi-UAV routing and scheduling problem for disaster-response missions in which a homogeneous fleet departs from a common depot to serve rescue points, each associated with a specific demand and a strict service time window, subject to route continuity, single-visit assignment, payload capacity, and UAV endurance constraints. Second, we introduce a load-aware energy consumption model in which consumption is a function of payload weight, capturing the realistic coupling between delivered load and flight energy requirements so that routing decisions better reflect true endurance limits during supply delivery missions. Third, multiple optimization strategies are implemented and evaluated under a unified framework to assess their effectiveness on the resulting large-scale combinatorial routing problem, compared through simulation studies across different mission scenarios using solution quality, convergence behavior, and computational efficiency as performance metrics.

The remainder of this paper is organized as follows. Section II presents the problem formulation, including decision variables, objective function, and constraints. Section III describes the optimization methodologies used to solve the problem. Section IV presents the simulation setup, experimental results, and discusses the obtained results and insights. Finally, Section V concludes the paper and outlines possible directions for future work.

II. PROBLEM FORMULATION

A. Problem Description

The optimization problem involves a set of UAVs $\mathcal{K} = \{1, 2, \dots, K\}$ delivering aid materials to a set of rescue points $\mathcal{N} = \{1, 2, \dots, N\}$. Each UAV departs from and returns to a common depot (node 0). UAVs are homogeneous, operating at a fixed speed v with identical payload and endurance limits. Each rescue point $i \in \mathcal{N}$ has a demand q_i and a service time window $[e_i, l_i]$. The travel time between any two nodes i and j is $t_{ij} = d_{ij}/v$, where d_{ij} is the Euclidean distance. Figure 1 represents an overview of the multi-UAV rescue network, where drones depart from a central depot to deliver supplies to multiple rescue points along their assigned routes, illustrating coordinated flight paths and task distribution.

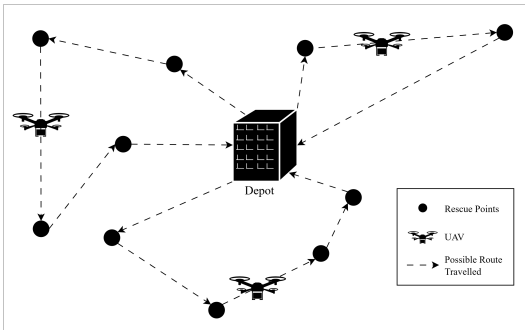


Fig. 1: Schematic Representation of UAVs Delivering Aid from a Central Depot to Multiple Rescue Points

B. Decision Variables

- $x_{ijk} = 1$ if UAV k travels from node i to node j , 0 otherwise.

- $o_k = 1$ if UAV k is activated, 0 otherwise.
- N_{ik} represents the remaining payload of UAV k after servicing node i .

C. Objective Function

The objective is to minimize total operational cost, including:

- Travel cost (proportional to distance),
- UAV activation cost (fixed cost per deployed UAV),
- Penalty cost for early or late arrivals relative to the time window.

The objective also seeks to maximize UAV efficiency by rewarding solutions that serve more points per active UAV and incentivizing timely arrivals. The combined objective function is:

$$\begin{aligned} \min C_{\text{total}} = & C_1 \sum_{k \in \mathcal{K}} \sum_{i \in \mathcal{N}_0} \sum_{j \in \mathcal{N}_0} t_{ij} x_{ijk} + C_2 \sum_{k \in \mathcal{K}} o_k + \\ & + \sum_{k \in \mathcal{K}} \sum_{i \in \mathcal{N}} (C_3 W_{ik} + C_4 L_{ik}) - C_5 \frac{\sum_{k \in \mathcal{K}} \sum_{i \in \mathcal{N}} y_{ik}}{\sum_{k \in \mathcal{K}} o_k} \\ & - C_6 \sum_{k \in \mathcal{K}} \sum_{i \in \mathcal{N}} \sum_{j \in \mathcal{N}} R_{ijk} \quad (1) \end{aligned}$$

where the auxiliary terms are defined as follows. $W_{ik} = \max(e_i - a_{ik}, 0)$ is the early-arrival waiting penalty for UAV k at node i , with a_{ik} denoting the arrival time and e_i the earliest service time. $L_{ik} = \max(a_{ik} - l_i, 0)$ is the late-arrival penalty, where l_i is the latest allowable service time. $y_{ik} \in \{0, 1\}$ is a binary indicator equal to 1 if UAV k serves node i , enabling the utilization-efficiency reward term. $R_{ijk} = x_{ijk} \cdot \mathbf{1}[a_{ik} \in [e_i, l_i]]$ rewards on-time traversals from node i to node j by UAV k , incentivising punctual arrivals. The cost coefficients $C_1, \dots, C_6$ weight travel cost, activation cost, early-arrival penalty, late-arrival penalty, utilization reward, and on-time traversal reward, respectively.

D. Constraints

Key constraints include:

- **Task assignment and flow conservation:** Each rescue point must be visited exactly once, and routes must be continuous:

$$\sum_{k \in \mathcal{K}} \sum_{j \in \mathcal{N}_0, j \neq i} x_{ijk} = 1, \quad \forall i \in \mathcal{N}. \quad (2)$$

- **UAV activation:** A UAV can only operate if it departs from the depot:

$$\sum_{j \in \mathcal{N}} x_{0jk} = o_k, \quad \forall k \in \mathcal{K}. \quad (3)$$

- **Payload and endurance constraints:** UAV payload and flight range are linked, and each UAV must respect its maximum distance based on current payload:

$$\sum_{i \in \mathcal{N}_0} \sum_{j \in \mathcal{N}_0} d_{ij} \left(1 + \alpha \frac{N_{ik}}{Q_{\max}} \right) x_{ijk} \leq D_{k0}, \quad \forall k \in \mathcal{K}. \quad (4)$$

- **Time-window and service constraints:** Service start times must respect travel times and time windows:

$$\begin{aligned} w_{jk} \geq & w_{ik} + s_i + t_{ij} \left(1 + \alpha \frac{N_{ik}}{Q_{\max}} \right) \\ & - M(1 - x_{ijk}), \quad \forall i, j \in \mathcal{N}_0, \forall k \in \mathcal{K}. \quad (5) \end{aligned}$$

These constraints ensure each UAV's task allocation, payload handling, endurance limits, and time window compliance, optimizing the overall cost while maximizing efficiency.

E. Assumptions and Limitations

UAVs are homogeneous, operating at fixed speed with identical payload and endurance limits. All rescue demands and time windows are known in advance, and the airspace is uncongested with no recharging mid-mission.

These assumptions can be relaxed in future extensions.

Fleet heterogeneity: UAV-specific Q_k and D_k parameters require only additional index sets over $\mathcal{K}$. **Stochastic environments:** wind disturbances or uncertain demands affect the load-dependent energy term (Eq. 3) and motivate robust or replanning-loop formulations. **No-fly zones** can be modeled as edge-weight penalties or graph pruning, while communication constraints motivate decentralized extensions.

III. METHODOLOGY

A. Baseline Metaheuristics

To provide comparative benchmarks, three classical metaheuristic algorithms were implemented: Simulated Annealing (SA), Genetic Algorithm (GA), and Particle Swarm Optimization (PSO).

Simulated Annealing (SA) explores the solution space by probabilistically accepting both improving and non-improving moves to escape local minima. The temperature schedule controls the exploration–exploitation trade-off [11]. For this study, parameters were set to $T_{\text{init}} = 2000$, $\alpha = 0.97$, $T_{\text{min}} = 0.01$, and $N_{\text{iter}} = 100$.

Genetic Algorithm (GA) evolves a population of routing solutions through selection, crossover, and mutation [12]. Feasibility is enforced by checking payload, endurance, and time-window constraints during evaluation. The population size and maximum generations were set to $P = 60$ and $G_{\text{max}} = 100$.

Particle Swarm Optimization (PSO) updates candidate routing solutions through velocity and position updates influenced by personal and global best solutions [13]. The effective segment cost accounts for load-dependent endurance via

$$d_{ij}^{\text{eff}} = d_{ij} \left(1 + \alpha \frac{N_k}{Q_{\text{max}}} \right).$$

Parameters were set to $w = 0.8$, $c_1 = 1.5$, $c_2 = 1.5$, $v_{\text{max}} = 0.3$, and $v_{\text{min}} = -0.3$.

B. Discrete Teaching–Learning-Based Optimization with Local Search (DTLBO-II-LS)

DTLBO-II-LS is proposed for combinatorial multi-UAV routing. Its three novelties beyond standard discrete TLBO are: (i) **Class-based multi-teacher learning** — the population is partitioned into sub-classes, each with its own teacher, promoting diversity. (ii) **Embedded local search** — swap, insertion, and 2-opt reversal operators refine solutions after each learning cycle. (iii) **Reconstruction mechanism** — when no improvement is detected across consecutive cycles, the worst-performing solutions are replaced with newly decoded ones, preventing stagnation. This combination is the key contributor to DTLBO-II-LS's

low run-to-run variance (std. 4.64 over 20 runs, Table I) and distinguishes it from a simple fusion of discrete TLBO with local search.

C. Policy-Gradient Reinforcement Learning (REINFORCE)

A policy-gradient RL framework based on REINFORCE is formulated for multi-UAV routing. The policy is parameterized by $\theta \in \mathbb{R}^N$; at each decision step, the next node is sampled from a softmax over the feasible (unvisited, constraint-compatible) set $\mathcal{F}$:

$$\pi_{\theta}(a | s) = \frac{\exp(\theta_a)}{\sum_{a' \in \mathcal{F}} \exp(\theta_{a'})}.$$

When no feasible node remains for the current UAV, it returns to depot and the next UAV activates. An episode terminates once all nodes are served or all UAVs have returned. Infeasible actions are masked throughout to guarantee constraint satisfaction.

The episodic reward is $r = -\text{cost}/1000$, aligning maximization with cost minimization. A moving-average baseline b is maintained; updates use advantage $(G - b)$ for variance reduction. Training uses learning rate $\eta = 0.01$ for 3,000 episodes per run, with hyperparameters chosen by grid search.

Scalability: at 20 nodes the sparse reward signal and flat θ parameterization struggle to capture long-range route interactions, causing a higher variance. Graph neural network encoders or actor–critic architectures are natural remedies explored in future work.

D. Framework Overview

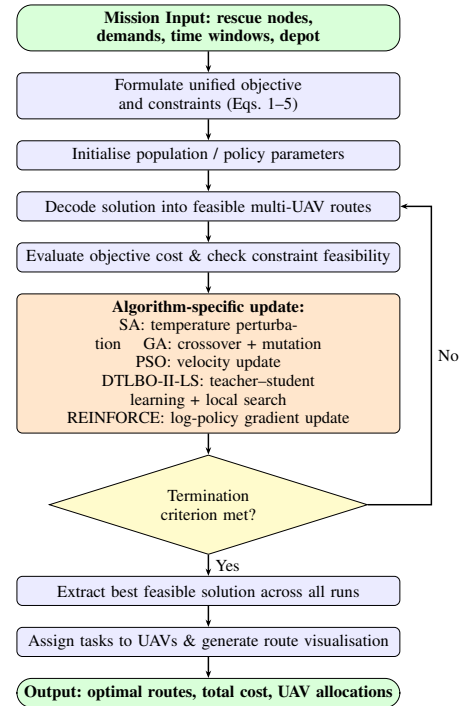


Fig. 2: Flowchart of the proposed multi-UAV routing framework. The orange block represents the algorithm-specific update step, which is the only component that varies across the five solvers. All other stages are shared by the unified framework.

Fig. 2 summarizes the end-to-end sequence of the proposed framework, from mission initialization through iterative optimization to final route assignment. The loop structure is shared by all five solvers; the algorithm-specific update block is the only step that differs across methods.

IV. SIMULATION AND RESULTS

All experiments reported in this section are computational simulations conducted in MATLAB R2024b; no physical UAV hardware or field flight trials were performed. Verification of the computed routes is carried out in two ways: (i) a constraint-satisfaction check, confirming that every generated route respects payload capacity, endurance, and time-window constraints, and (ii) benchmark replication, in which the exact node coordinates, demands, and time windows from a published instance (Han et al.) are reproduced and the resulting objective costs are compared against their reported result (Section IV-C). Five optimisation algorithms are benchmarked: Simulated Annealing (SA), Genetic Algorithm (GA), Particle Swarm Optimisation (PSO), Discrete Teaching-Learning-Based Optimisation with local search (DTLBO-II-LS), and the REINFORCE policy-gradient method. Two scenarios are considered:

- **Case 1:** $N = 10$ rescue nodes, $Q_{\max} = 50$ kg.
- **Case 2:** $N = 20$ rescue nodes, $Q_{\max} = 50$ kg.

Across both cases, UAV speed is $v = 15$ m/s, empty-range $D_{k0} = 150$ km, and the endurance-payload coupling coefficient is $\alpha = 0.3$. Cost coefficients are set to $(C_1, \dots, C_6) = (1000, 100, 800, 1000, 50, 200)$. For additional validation, a published benchmark from Han et al. ($N = 10$, $Q_{\max} = 20$ kg) is reproduced using the original node coordinates, demands, and time windows.

A. Case 1: 10 Rescue Nodes

Figure 3 shows the best objective cost achieved by each algorithm in Case 1. All five methods deploy three UAVs, enabled by the large payload capacity which consolidates deliveries into fewer, longer routes.

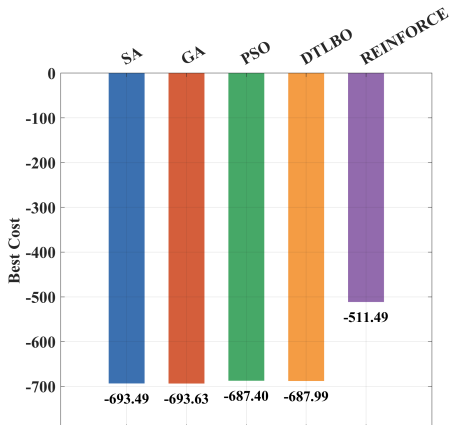


Fig. 3: Best objective cost for each algorithm in Case 1 ($N = 10$, $Q_{\max} = 50$ kg, 3 UAVs active). Lower values indicate better solutions.

SA and GA achieve near-identical best costs (-693.49 and -693.63 , respectively), making them the strongest performers in this scenario. PSO (-687.40) and DTLBO-II-LS (-687.99) trail marginally, while REINFORCE lags

significantly at -511.49 , reflecting limitations in its current reward structure and training configuration under large-payload conditions.

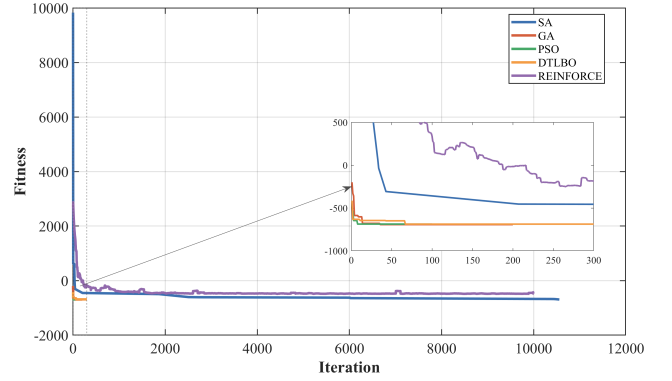


Fig. 4: Elite-run convergence curves in Case 1. Inset shows a magnified view of the early iterations where SA, GA, PSO, and DTLBO-II-LS separate from REINFORCE.

Fig. 4 shows best-so-far cost vs. iteration. All meta-heuristics descend quickly; SA and GA improve most steeply. REINFORCE begins in the thousands and converges slowly, confirming the difficulty of policy-gradient training here.

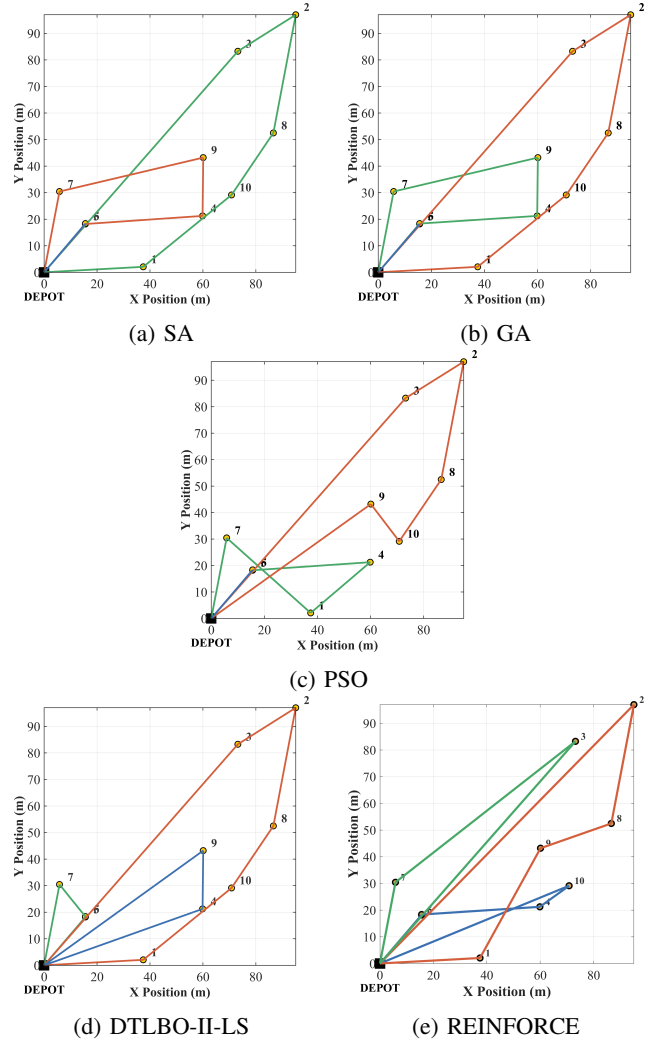


Fig. 5: Optimal UAV routing solutions for Case 1. Colours represent distinct UAVs; squares = depot, circles = rescue nodes.

All routes in Fig. 5 respect endurance, capacity, and time-window constraints. SA and GA yield the most compact, balanced partitions consistent with their superior costs.

B. Case 2: 20 Rescue Nodes

Figure 6 compares algorithm performance in Case 2, where doubling the node count to 20 raises problem complexity considerably and requires six UAVs across all methods.

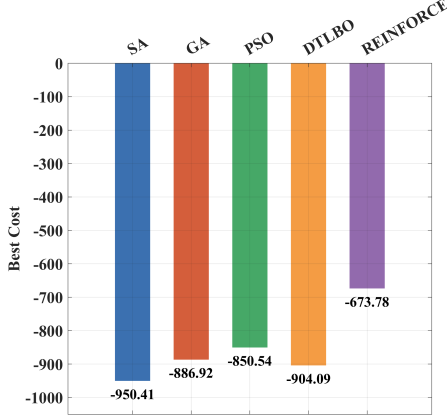


Fig. 6: Best objective cost for each algorithm in Case 2 ($N = 20$, $Q_{\max} = 50$ kg, 6 UAVs active). Lower values indicate better solutions.

SA delivers the best solution (-950.41), followed by DTLBO-II-LS (-904.09), GA (-886.92), and PSO (-850.54). REINFORCE lags at -673.78 , confirming sensitivity to problem scale.

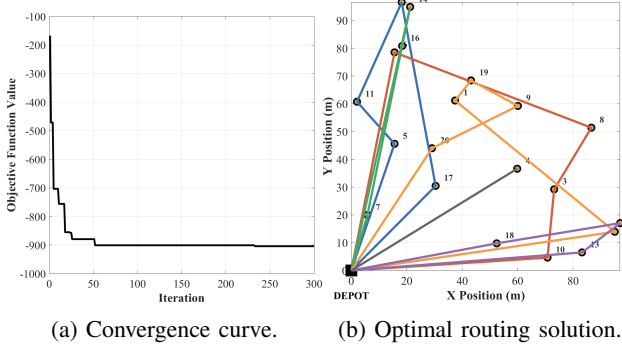


Fig. 7: DTLBO-II-LS results for Case 2

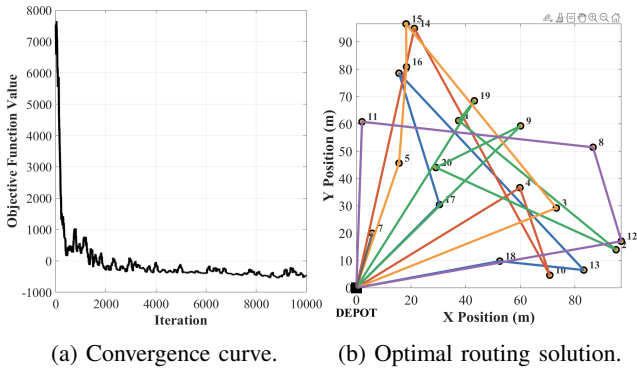


Fig. 8: REINFORCE results for Case 2

Figs. 7 and 8 show that DTLBO-II-LS converges steadily with balanced load distribution, while REIN-

FORCE exhibits slower convergence and more irregular routing in the larger action space.

C. Benchmark Validation

On the Han et al. benchmark ($N = 10$, $Q_{\max} = 20$ kg), the published PSOGWO cost is 67.6052 with seven UAVs. All five solvers yield more negative costs with only three to five UAVs. Cost magnitudes differ because our objective includes reward terms (C_5 , C_6) producing negative costs with different coefficient scaling. The validation confirms that (i) all constraints are satisfied, (ii) fewer UAVs are activated, and (iii) all solvers are mutually consistent on identical input data.

D. Limitations

The present evaluation has several limitations. Validation is simulation-based only; no hardware or field UAV trials were conducted. Only two problem sizes were tested ($N = 10$ and $N = 20$); scalability to substantially larger rescue networks remains unverified. REINFORCE uses a flat softmax policy without a learned state encoder (e.g., a graph neural network or actor-critic architecture), which likely understates the achievable performance of learning-based approaches at scale. The simulated environment is static and uncongested, with no obstacles, no-fly zones, weather effects, or communication constraints, and the UAV fleet is treated as homogeneous throughout. Addressing these points — heterogeneous fleets, stochastic or dynamic environments, no-fly-zone modelling, and richer RL architectures — is left to future work (Section V).

E. Multi-Run Statistical Comparison

To assess robustness, each algorithm is executed over 20 independent runs on Case 1. Table I summarises elite cost, worst cost, mean cost, standard deviation, and mean run-time. GA achieves the strongest elite solution (-707.71); DTLBO-II-LS provides the best mean (-673.41) and lowest std. (4.64), indicating high run-to-run consistency. PSO is fastest at 0.17 s/run, making it attractive for rapid replanning. REINFORCE, though fastest overall, yields the weakest mean and highest variance (30.55), reflecting sensitivity to reward scaling and sparse feasible solutions.

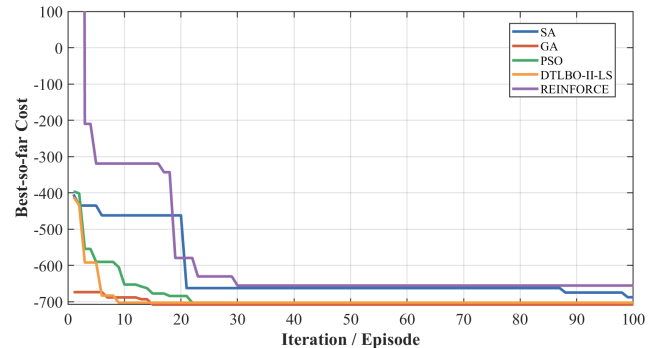


Fig. 9: Elite-run convergence comparison for all five algorithms over 20 runs, Case 1 ($N = 10$, $Q_{\max} = 50$ kg).

Figure 9 shows the zoomed convergence comparison (y -axis clipped at $+100$). GA and DTLBO-II-LS descend fastest; SA converges steadily; PSO oscillates mildly. REINFORCE converges slowest, consistent with its elevated standard deviation.

TABLE I: Statistical comparison over 20 independent runs for Case 1 ($N = 10$, $Q_{\max} = 50$ kg). The **elite cost** is the best single-run result; lower cost is better for all cost metrics.

Metric	SA	GA	PSO	DTLBO-II-LS	REINFORCE
Elite cost	−687.40	−707.71	−702.41	−702.41	−655.31
Worst cost	−603.04	−587.94	−520.54	−614.46	−150.30
Mean cost	−661.43	−652.52	−628.28	−673.41	−379.84
Std. deviation	5.27	8.87	11.21	4.64	30.55
Mean runtime [s]	0.34	0.37	0.17	2.40	0.10

V. CONCLUSION AND FUTURE WORK

This paper presented a constrained multi-UAV routing framework for disaster rescue operations with load-dependent endurance coupling and time-window constraints. Five optimization strategies were evaluated across 10- and 20-node scenarios. SA and GA achieve strong single-run quality; DTLBO-II-LS delivers the best statistical robustness (std. 4.64 over 20 runs) through its multi-class learning, embedded local search, and stagnation-recovery mechanism; PSO offers the fastest runtime (0.17 s/run); and REINFORCE provides operator-free adaptive routing at the cost of higher variance at scale.

These findings position structured discrete metaheuristics as deployment-ready optimizers and RL as a flexible long-term learning direction—complementary paradigms rather than competing ones. Future work will address heterogeneous fleets, stochastic environments, graph-neural-network policy encoders, actor–critic architectures, and online replanning for field deployment.

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